

114 學年度第二學期 離散數學 HW05

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1 題目

1.1 作業繳交說明

- 從 overleaf 下載 hw05.zip
- 從 overleaf 下載 hw05.pdf
- 建立一個新檔案夾，命名如下
- 離散數學_114590xxx_姓名_HW05
- 放入 hw05.zip
- 放入 hw05.pdf
- 將此檔案夾，壓縮成 zip 檔，檔名如下
- 離散數學_114590xxx_姓名_HW05.zip
- 將此壓縮檔，上傳到北科 i 學園 PLUS
- Do not share your homework with any living creatures.
- 作業遲交以零分計。

1.2 題號

- page 350, chapter 5.1 Exercise 6
- page 351, chapter 5.1 Exercise 18

2 作答

2.1 page 350, chapter 5.1 Exercise 6

Prove that $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = (n+1)! - 1$ whenever n is a positive integer.

Prove $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = (n+1)! - 1$

For all $n \geq 1$, $P(n) = (n+1)! - 1$

Basis step:

$n = 1$, $P(1)$ is true, since $1 \cdot 1! = (1+1)! - 1 = (2)! - 1$

Inductive step:

Inductive hypothesis, assume $P(k)$ holds for an arbitrary positive

integer k . $1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! = (k+1)! - 1$

Add $(k+1) \cdot (k+1)!$ to both sides of the equation in $P(k)$, we obtain

$$\begin{aligned} 1 \cdot 1! + \dots + k \cdot k! + (k+1) \cdot (k+1)! &= (k+1)! - 1 + (k+1) \cdot (k+1)! \\ &= (k+1)!(1 + k + 1) - 1 \\ &= (k+2)! - 1 \end{aligned}$$

By mathematical induction, $P(n)$ is true for all integer n with $n \geq 1$.

2.2 page 351, chapter 5.1 Exercise 18

Let $P(n)$ be the statement that $n! < n^n$, where n is an integer greater than 1.

- What is the statement $P(2)$?
- Show that $P(2)$ is true, completing the basis step of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1.

- What is the inductive hypothesis of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1?
- What do you need to prove in the inductive step of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1?
- Complete the inductive step of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1.
- Explain why these steps show that this inequality is true whenever n is an integer greater than 1.

- Plugging in $n = 2$, we see that $P(2)$ is the statement $2! < 2^2$.
- Since $2! = 2$, this is the true statement $2 < 4$.
- The inductive hypothesis is the statement that $k! < k^k$.
- For the inductive step, we want to show for each $k \geq 2$ that $P(k)$ implies $P(k+1)$. In other words, we want to show that assuming the inductive hypothesis (see part (c)) we can prove that $(k+1)! < (k+1)^{k+1}$.
- $(k+1)! = (k+1) \cdot k! < (k+1) \cdot k^k < (k+1) \cdot (k+1)^k = (k+1)^{k+1}$
- We have completed both the basis step and the inductive step, so by the principle of mathematical induction, the statement is true for every positive integer n greater than 1.